ARGO ANALYTICS · PROPOSAL

Lee Enterprises

Conversational AI & Canonical Data Platform

Prepared by **Argo Analytics** · Version 1 · May 2026

Executive Summary

Lee has the right cloud foundation in place — BigQuery, MWAA/Airflow, dbt, and BI tooling through Looker and Domo — but the layers that turn raw data into governed, reusable assets are still emerging. There is no established medallion pattern, no canonical semantic model, and no governed path for AI-assisted analytics. This proposal establishes that framework, using subscription rates as the initial dataset to prove the pattern end-to-end.

This proposal lays out a phased program that turns Lee's existing BigQuery and dbt investment into a canonical, AI-ready data foundation. Phase 1 establishes a reusable subscription rates model and a governed conversational interface on top of it. Subsequent acquisitions onboard against that model instead of rebuilding from scratch.

Three outcomes drive the work:

- **Foundational pattern.** Establish the medallion architecture, canonical modeling approach, and governance pattern that all future domains and acquisitions will follow.
- **Analyst leverage.** Move routine subscription rates questions from "ticket the BI team" to natural-language self-serve, backed by governed SQL execution against trusted models.
- **Acquisition leverage.** Position Lee to onboard the next acquisition against a canonical model rather than rebuilding pipelines and KPIs from scratch.

> Phase 1 delivers a governed conversational AI experience over a canonical subscription rates model — proving the pattern on a high-value dataset before scaling. Phases 2 and 3 are directional; they activate only after Phase 1 validates adoption and ROI.

Strategic Context

What this proposal establishes

This proposal establishes four reusable patterns:

- **Medallion layering in BigQuery.** Raw → staging → intermediate → mart → AI-semantic, with consistent conventions across domains.
- **Canonical semantic modeling.** Source-agnostic dbt models defined once, mapped per source system. New sources map into the model rather than producing parallel ones.
- **Governed AI orchestration.** Context assembly, SQL generation, dry-run validation, scoped execution, and logging — the pattern that makes conversational AI safe for enterprise data.
- **Repeatable acquisition onboarding.** A documented playbook for mapping a new property's source data into the canonical layer and exposing it through the same governed consumption paths.

Phase 1 builds all four against the subscription rates dataset. Every subsequent domain and acquisition reuses the patterns rather than reinventing them.

Current state

Lee operates a centralized cloud data platform with the right ingredients:

- AWS MWAA / Airflow for orchestration.
- GCS as a landing zone for source extracts.
- BigQuery as the analytical warehouse.
- dbt for transformation, with growing adoption.
- Looker and Domo for BI consumption.

The architecture is sound at the foundation. The gap is at the upper layers: there is no established medallion pattern, the dbt project is early in its maturity, semantic definitions are inconsistent across sources, and there is no governed path for AI-assisted analytics. Phase 1 closes those gaps for the subscription rates dataset and produces the framework that every future domain follows.

Proposed Solution

A phased program that builds the canonical data foundation and the AI layer together..

Architecture overview

The solution establishes three coordinated layers that together define Lee's analytics framework going forward:

- **Canonical semantic models.** Source-agnostic dbt models that consolidate per-property subscription rate data into a single shared definition. New acquisitions map into this model rather than producing parallel ones.
- **AI-ready semantic layer.** A purpose-built downstream layer in BigQuery designed for LLM consumption — denormalized where helpful, annotated with descriptions, paired with markdown context files that ground generated SQL.
- **Conversational AI orchestration.** An orchestration service that assembles context, prompts Gemini via Vertex AI, validates generated SQL with a BigQuery dry-run, executes against governed models, and returns grounded answers.

Three consumption flows, one foundation

The platform supports three ways to consume governed data, all reading from the same models:

- **Dashboards.** Looker and Domo continue to serve curated reporting against gold models. No regression of existing capability.
- **Direct SQL.** Analysts query gold and AI-semantic models directly through their existing SQL IDE.
- **Conversational AI.** Business users ask questions in natural language and receive grounded, source-cited answers backed by governed SQL execution.

Phase 1 — Subscription Rates Canonical Model + MVP Conversational AI

Scope

Phase 1 covers a single business dataset (subscription rates) end-to-end: from raw source data through canonical and AI-ready models, into a conversational interface with a prototype Teams-based or custom chatbot-style UI.

In scope

- **Canonical subscription rates model.** Discovery, definition, and dbt build covering at least Falcon and DSI sources. Mapping pattern documented for future acquisitions.
- **Gold + AI-semantic build-out.** Production-grade gold subscription rates model and downstream AI-semantic model with annotations and descriptions.
- **Markdown context framework.** Versioned context files covering business definitions, synonyms, KPI logic, example questions, and acquisition-specific nuance. Stored in the dbt repo.
- **AI orchestration service.** Backend service that assembles context, calls Gemini via Vertex AI, validates generated SQL via BigQuery dry-run, executes, and returns grounded responses.
- **Prototype Teams or custom chatbot-style UI.** A lightweight conversational interface sufficient for validation, demo, and pilot rollout to a defined user group. Not a production-hardened enterprise app.
- **Governance + access controls.** IAM-based scoping of which models the AI can query, plus logging of all generated SQL and executions for audit.
- **Documentation + handoff.** Architecture docs, runbooks, onboarding pattern for new acquisitions, context-file authoring guide.

Out of scope

- Additional business domains (advertising, audience, editorial) — addressable in follow-on phases.
- Production-grade Teams app (full enterprise distribution, SSO hardening, full admin tooling).
- dbt Semantic Layer API integration — Phase 2.
- Dataplex metadata integration — Phase 2.
- Persona-based autonomous agents — Phase 3.
- Vector knowledge graph / RAG over unstructured content.

Acceptance criteria

Phase 1 is complete when:

- The canonical subscription rates model is in production, sourced from Falcon and DSI, with documented mapping logic.

- A defined set of pilot questions (agreed with Lee during discovery) can be asked in natural language and answered with grounded, accurate SQL execution.
- Governance controls restrict AI execution to approved models, and all executions are logged.
- Acquisition onboarding pattern is documented and reviewed with Lee leadership.
- Argo conducts a handoff and enablement session covering operations, context-file authoring, and runbook walk-through.

Phases 2 & 3 — Directional

Phases 2 and 3 are illustrative, not committed. They activate only after Phase 1 demonstrates adoption and ROI, and they are scoped through separate SOWs.

Phase 2 — Semantic & governance expansion

Adds depth to the foundation:

- Integrate the dbt Semantic Layer for governed metric contracts shared across BI and AI consumption.
- Integrate Dataplex for metadata, lineage, and policy-aware context assembly.
- Expand the canonical model to additional domains (advertising, audience).
- Onboard the first real acquisition end-to-end using the documented pattern.

Phase 3 — Agentic AI layer

Once the semantic foundation is mature, introduce specialized agents:

- Persona agents (subscriber analytics, revenue ops, acquisition integration) with scoped tool access.
- Multi-step reasoning and cross-system workflow execution.
- Proactive insight generation against governed data.

Phase 3 deliberately follows the governance work in Phase 2. Agents without strong semantic grounding produce confident wrong answers, which erodes trust faster than no AI at all.

Phase 1 — Level of Effort

Phase 1 — Investment

Argo proposes a **fixed-bid engagement of $20,000** for the scope defined above.

The estimated underlying effort is ~150 hours. Argo is investing the difference to support a clean launch of the platform pattern and to align incentives around getting the foundation right the first time.

Fixed scope includes everything in the *In Scope* list above. Items in the *Out of Scope* list, or new requests that emerge during the engagement, are addressed through a separate change order at the standard $200/hr rate.

GCP usage costs (Vertex AI / Gemini, BigQuery) are usage-based, separate from this engagement, and billed directly by Google to Lee. At pilot scale these are expected to be modest — low hundreds to low thousands per month.

Billing: 50% at engagement kickoff, 50% at delivery and handoff acceptance.

Why fixed-bid:

- Removes scope uncertainty for Lee on an AI-enabled engagement
- Establishes a clean foundation for follow-on platform work

Assumptions

- BigQuery, MWAA/Airflow, dbt, Looker, and Domo remain the foundational stack.
- GCP Vertex AI / Gemini is the AI provider. No cross-cloud LLM evaluation in Phase 1.
- The prototype uses Microsoft Teams or a lightweight custom chatbot-style UI as the conversational entry point.
- KPI reconciliation requires named Lee stakeholders with authority to ratify canonical definitions when Falcon and DSI disagree.
- Subscription Rates is the Phase 1 dataset. Other domains follow in subsequent phases.

Risks & Mitigations

Risk	Impact	Mitigation
KPI reconciliation across Falcon and DSI stalls	Schedule slip; canonical model blocked	Named Lee owner per KPI; time-boxed decisions; document deltas explicitly
LLM hallucination on edge-case questions	Erodes user trust early	Dry-run validation; constrained schema scope; "I don't know" routing; logging + review
Pilot user adoption is thin	Hard to validate ROI	Pick pilot users with real recurring subscription rate questions; track resolved questions weekly
Source data gaps surface mid-build	Rework in gold layer	Schema and sample-data review in discovery before canonical model design is finalized

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Next Steps

- Lee review of architecture and phased roadmap.
- Confirm Phase 1 scope and acceptance criteria with named Lee stakeholders.
- Identify the pilot user group and 10–15 seed questions to anchor discovery.
- Confirm Falcon and DSI as the Phase 1 source systems and assign data owners.
- Kickoff discovery sprint (week-by-week plan provided at SOW execution).